جامعة بيرزيت

BIRZEIT UNIVERSITY

Data Science (COMP4381)

Project Second Phase

Flight Delay Analysis Using Data Science Techniques

Instructor: Ahmed Sabbah

May 2, 2026

Team members	Id
Tala khateeb	1222091
Bara Daraj	1230667
Yosra Mahmoud	1200867

Abstract:

Flight delays represent a major challenge in the aviation industry due to their impact on passengers, airline operations, and overall system efficiency. These delays are influenced by a variety of factors, with weather conditions being one of the most significant contributors. This project aims to analyze the effect of weather and other operational factors on flight delays using aviation data.

The study focuses on identifying key patterns and relationships between variables such as visibility, wind speed, precipitation, and air traffic conditions. By applying data analysis techniques and considering machine learning approaches used in recent studies, the project seeks to better understand how these factors contribute to delays and how they interact with each other.

The findings of this project are expected to provide insights that can help improve decision-making in airline operations and contribute to reducing delays. Ultimately, this can enhance passenger experience and increase the efficiency of the aviation system.

Introduction:

Flight delays are one of the most persistent challenges in the aviation industry, affecting millions of passengers and causing significant operational and financial impacts on airlines and airports. A flight delay occurs when an aircraft departs or arrives later than its scheduled time. These delays can result from multiple interconnected factors, including weather conditions, air traffic congestion, technical malfunctions, and operational constraints.

Among all contributing factors, weather conditions play a critical role in disrupting flight operations. Severe weather events such as thunderstorms, heavy rain, fog, strong winds, and low visibility can directly affect airport safety and efficiency. In many cases, weather disturbances do not only impact a single flight but can also create a cascading effect across multiple airports and flight routes.

Understanding the causes of flight delays is essential for improving airline performance, enhancing passenger satisfaction, and optimizing airport operations. Therefore, this project aims to analyze the impact of weather and other influencing factors on flight delays in order to better understand delay patterns and their underlying causes.

Background:

Flight delays are a common and complex issue in modern air transportation systems. A flight delay is typically defined as the difference between the scheduled time and the actual time of departure or arrival. Delays can occur at various stages of a flight, including departure delays at the origin airport, en-route delays during the flight, and arrival delays at the destination.

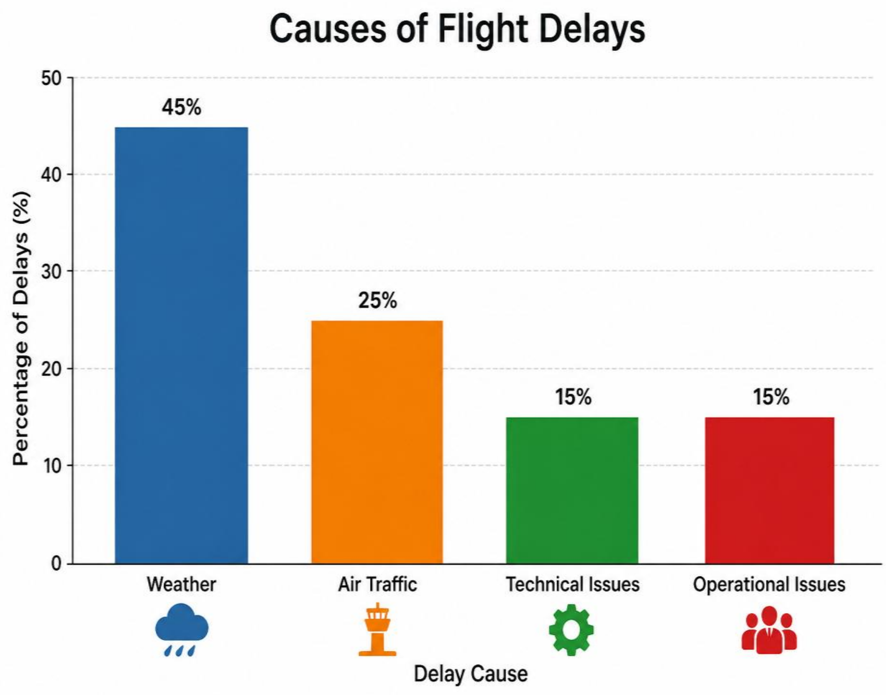
The causes of flight delays are diverse and often interconnected. These causes can generally be categorized into weather-related factors and non-weather-related factors. Weather conditions are among the most critical factors, as they directly impact flight safety and operational efficiency. Conditions such as heavy rain, thunderstorms, fog, strong winds, and low visibility can limit aircraft movement, reduce runway capacity, and require additional safety measures. In severe cases, flights may be delayed for extended periods or even canceled.

In addition to weather, several operational and technical factors contribute to delays. Air traffic congestion is a major issue, especially in busy airports where the number of scheduled flights exceeds the available capacity. This leads to longer waiting times for takeoff and landing. Airport infrastructure limitations, such as the number of runways and gates, also affect how efficiently flights can be handled.

Technical problems, including aircraft maintenance issues, can cause unexpected delays as safety inspections and repairs must be completed before departure. Furthermore, airline-related factors such as crew availability, scheduling decisions, and turnaround time between flights also play a role in determining whether a flight departs on time.

Another important concept is delay propagation, where a delay in one flight affects subsequent flights. Since aircraft are often scheduled for multiple trips throughout the day, a delay in an earlier flight can lead to a chain reaction, causing additional delays across different routes and airports.

Understanding these background concepts is essential for analyzing flight delays effectively. It provides a foundation for examining how different factors—especially weather conditions—interact and influence flight performance. This understanding supports the development of data-driven approaches to identify patterns and improve operational decision-making in the aviation industry.



Domain explanation:

This project is classified under the aviation data analytics domain, which focuses on analyzing and interpreting aviation-related data to improve decision-making and operational efficiency within the airline industry.

The aviation sector generates large-scale datasets daily, including flight schedules, actual departure and arrival times, airport conditions, and weather reports. By analyzing this data, researchers can identify patterns, correlations, and key factors that contribute to flight delays.

A major focus within this domain is understanding the relationship between weather conditions and flight performance. Weather variables such as precipitation, wind speed, fog intensity, and storms are known to significantly influence flight punctuality and safety. In addition to weather, other operational factors such as air traffic volume, airport capacity, and airline scheduling strategies also contribute to delays.

The use of data analysis techniques in this domain allows researchers and aviation authorities to develop predictive models that can forecast delays and support better planning. Ultimately, this helps improve efficiency, reduce costs, and enhance the overall passenger experience in the aviation system.

Algorithm background:

To analyze the impact of weather and other factors on flight delays, this project relies on data analysis and machine learning techniques. These approaches allow for identifying patterns within large datasets and understanding how different variables influence flight delay behavior.

One of the fundamental approaches used in this context is regression analysis. Regression models are designed to predict continuous values, such as the duration of a flight delay. By using input features like wind speed, temperature, visibility, precipitation, and air traffic conditions, regression models can estimate how much delay is expected under certain conditions. Linear Regression is one of the simplest methods, providing a clear understanding of the relationship between variables.

In addition to regression, classification algorithms can also be applied. Instead of predicting the exact delay time, classification methods group flights into categories such as “on-time,” “slightly delayed,” or “heavily delayed.” Decision Trees are commonly used because they are easy to interpret and can clearly show which factors have the greatest impact on delays. Random Forest, which is an extension of Decision Trees, improves prediction accuracy by combining multiple trees and reducing overfitting.

Another important aspect of the analysis is feature importance, which helps identify the most influential variables affecting flight delays. For example, machine learning models can determine whether visibility, wind speed, or precipitation has a stronger effect on delays under different conditions.

Before applying these algorithms, data preprocessing steps are usually required. These include handling missing values, normalizing data, and selecting relevant features. Proper preprocessing improves the accuracy and reliability of the models.

By applying these techniques, the project aims not only to analyze past flight delay data but also to support predictive modeling. This can help airlines and airport authorities anticipate delays and take proactive measures to reduce their impact, ultimately improving efficiency and passenger experience.

Related Work:

- In a study conducted in September 2022, researchers investigated the impact of space weather on flight delays by analyzing a large dataset of approximately 5×10^6 flight records. The study found that during space weather events, the average arrival delay time increased significantly, while the rate of delays exceeding 30 minutes increased by about 21.45%. In addition, the overall delay time showed an increase of approximately 81.34% compared to normal conditions. The results indicate that disturbances in communication and navigation systems caused by solar activity contribute to these delays. These findings are relevant to our project as they highlight how environmental factors can significantly affect flight performance and scheduling *(Y. Wang et al., 2022).*

- Results from a study published in March 2025 showed that visibility has a significant impact on flight delays and disruptions. By analyzing historical flight and weather data from Yakubu Gowon Airport in Jos, Nigeria, researchers found that poor visibility conditions such as fog, thunderstorms, and haze greatly increase the likelihood of delays and cancellations. Statistical analysis revealed a moderate positive correlation ($R = 0.530$) between visibility and flight delays, with approximately 28% of the variation in delays attributed to visibility factors. The study also highlighted the importance of advanced technologies such as Instrument Landing Systems (ILS) in reducing visibility-related disruptions. These findings are relevant to our project as they demonstrate how reduced visibility directly affects flight performance and scheduling *(Akintunde et al., 2025).*

- In a recent study, researchers proposed a novel model called MoEformer for predicting flight delays using a Mixture-of-Experts framework. The model uses a dynamic mechanism to activate specialized components based on different flight conditions, improving prediction accuracy and flexibility. The results showed that the proposed model outperforms existing deep learning approaches and maintains strong performance even in new flight scenarios without requiring full retraining. These findings are relevant to our project as they demonstrate advanced machine learning techniques for improving flight delay prediction *(Tian et al., 2026)*

- In a recent study, researchers proposed a modified Random Forest model to improve flight delay prediction by considering the priority of weather and non-weather features. The study analyzed U.S. domestic flight data and applied clustering techniques to examine the impact of different factors on delays. The results showed that weather features play a dominant role in severe conditions, leading to longer delays, while non-weather features contribute more to shorter delays under normal conditions. The proposed model significantly improved prediction accuracy compared to traditional methods. These findings are relevant to our project as they highlight the importance of combining different feature types to enhance flight delay prediction. (Li et al., 2023)

- In a study conducted between 2012 and 2019 at Kuwait International Airport, researchers examined the impact of weather conditions on flight departure delays. The study applied several machine learning algorithms, including Logistic Regression, Decision Tree, Neural Network, and Ensemble Learning, on a dataset combining flight and weather information. The results showed that dust storms and precipitation were the most influential factors affecting delays. Among the models, the Ensemble method achieved the best performance with an accuracy of 70.5%, a true positive rate of 82.6%, and an F1-score of 77.3%. These findings are relevant to our project as they demonstrate how weather-related factors can be effectively used with machine learning techniques to predict flight delays. *(Alkheder et al., 2026)*

Study	Year	Data Size	Method	Key Result
Wang et al.	2022	5×10^6	Statistical Analysis	Delay ↑ 81%
Akintunde et al.	2025	Historical Data	Correlation	28% due to visibility
Tian et al.	2026	Real-world datasets	MoEformer (ML)	Best performance
Li et al.	2023	U.S. flight data	Random Forest (ML)	Accuracy improved
Alkheder et al.	2026	Kuwait dataset	ML Models	Accuracy 70.5%

Contents